

Chapter 1

INTRODUCTION

An automatic cattle feeding bot is a smart agricultural robot designed to feed livestock with minimal human intervention. It helps farmers improve feeding efficiency, reduce labor costs, and ensure animals receive food at the right time and in the correct quantity. Agriculture and dairy farming are among the most important sectors for food production and economic development. In dairy farms, proper feeding management is essential to maintain cattle health, improve milk production, and increase overall farm efficiency. Traditional cattle feeding methods mainly depend on manual labor, where farmers are required to feed cattle regularly and monitor feeding operations continuously [1].

This process becomes difficult and time-consuming especially in farms with a large number of cattle. Irregular feeding schedules and improper feed management may negatively affect cattle growth, productivity, and livestock health. Therefore, there is a growing need for intelligent and automated systems that can simplify cattle feeding operations and reduce human effort. Recent studies on IoT-based livestock feeding systems and automated feeding technologies have highlighted the importance of smart monitoring and precision feeding in modern dairy farming [2].

Recent advancements in robotics, computer vision, IoT, and embedded systems have enabled the development of smart farming technologies for modern livestock management. Automated cattle feeding systems use intelligent image processing and robotic automation to identify cattle and dispense feed automatically without continuous human supervision[3].

Computer vision techniques combined with artificial intelligence models can accurately detect cattle in real time using camera input, making automated feeding systems more efficient and reliable. Researchers have also explored automated monitoring of cattle feeding behaviour using sensors and accelerometer-based systems to improve livestock management and feeding analysis. In addition, wireless monitoring and smart sensing technologies have been integrated into cattle monitoring systems to improve automation and energy efficiency in farm environments[4].

The proposed automatic cattle feeding bot is designed to automate cattle feeding operations using raspberry Pi 5, TensorFlow Lite AI model, OpenCV image processing, USB web camera, L298N motor driver, DC motors, and servo-controlled feed dispensing mechanism. The USB web camera continuously captures real-time video frames from the surrounding environment and transfers them to the Raspberry Pi for image processing [5].

The captured images are analyzed using Open CV and a TensorFlow Lite image classification model trained using Google Teachable Machine. When cattle are detected successfully with sufficient confidence, the robot automatically stops near the cattle and activates the feed dispensing mechanism using a servo motor [6].

Feed is dispensed in controlled quantity, after which the robot resumes autonomous movement and continues monitoring the environment. Similar automated feed dispensing mechanisms and embedded controller-based livestock feeding systems have been proposed in earlier research works for improving feeding accuracy and reducing human effort [7].

The proposed system combines image processing, embedded systems, and robotic automation technologies to create an intelligent livestock feeding solution. The robot operates autonomously inside the farm environment and reduces the need for manual cattle feeding operations [8].

The integration of camera-based cattle detection with autonomous robotic movement improves feeding consistency, reduces feed wastage, and enhances operational efficiency in dairy farms. Precision feeding approaches based on animal performance and automated sensing systems have also demonstrated positive impacts on dairy production efficiency and livestock health management [9].

The automatic cattle feeding bot is designed as a portable, cost-effective, and efficient smart farming solution suitable for modern livestock management applications. The system demonstrates the practical implementation of TensorFlow Lite AI models on embedded devices such as Raspberry Pi for real-time agricultural automation [10].

1.1 Motivation

Traditional cattle feeding methods require continuous human effort and regular monitoring, making the process labor-intensive and time-consuming. In many dairy farms, farmers face difficulties maintaining proper feeding schedules due to workload, lack of manpower, and increasing farm size. Irregular feeding can negatively affect cattle health, growth, and milk production. Manual feeding also increases the chances of feed wastage and inconsistent feed distribution.

With the rapid advancement of automation, robotics, and IoT technologies, there is a growing need for smart agricultural solutions that can improve farm productivity while reducing human effort. Automated feeding systems can help farmers manage livestock more efficiently and ensure timely feed delivery without constant supervision.

The motivation behind the automatic cattle feeding bot project is to develop an intelligent and autonomous system that can detect cattle in real time and dispense feed automatically. The project aims to combine Camera-based computer vision with robotic automation to improve feeding efficiency, reduce labor dependency, minimize feed wastage, and support modern smart farming practices.

1.2 Objectives

The automatic cattle feeding bot project aims towards achieving the following core goals.

1. To develop an autonomous robotic system capable of dispensing feed automatically to cattle without continuous human intervention, thereby reducing manual labor involved in livestock feeding operation.
2. To implement a camera based cattle detection system using TensorFlow Lite and OpenCV image processing techniques for real-time identification of cattle inside the farm environment.
3. To design a robotic platform capable of moving autonomously within the farm using DC motors and motor driver control for continuous cattle monitoring and feeding operations.
4. To develop a servo motor based feed dispensing mechanism that releases feed in controlled quantity whenever cattle are detected successfully by the artificial intelligence model.

Chapter 2

LITERATURE SURVEY

The literature survey is included in the report to review existing research works, understand current developments, and identify the importance of automation technologies in livestock farming systems. It helps establish a foundation for the proposed work by analyzing previous studies, comparing methodologies, and identifying the need for intelligent and autonomous cattle feeding systems using image processing, and embedded technologies.

N. Umapathi et al. presented a IoT-based automated feeding system for livestock management. The system uses sensors, microcontrollers, and wireless communication technologies to monitor feed levels and automate feeding schedules. The proposed model reduces manual labor and improves feeding efficiency by enabling real-time monitoring and control through IoT platforms. The work mainly focuses on improving operational efficiency and minimizing feed wastage in livestock farms [1].

C. Giannone et al. presented a computer vision-based monitoring system for dairy cattle. The paper explains how camera based image processing techniques can continuously observe cattle behaviour such as feeding activity, movement, and health conditions. The system provides non-contact monitoring and helps in identifying abnormal behaviour in real time, thereby improving dairy farm management and animal welfare [2].

V. C. Souza et al. presented precision feeding techniques where dairy cows receive customized feed according to their historical production performance. The research analyzed the effects of short-term supplementation changes on milk production and feed efficiency. The study demonstrates that data-driven feeding strategies can optimize nutrition and improve dairy productivity while reducing unnecessary feed usage [3].

J. Tinoco et al. developed a multi-sensory cattle monitoring system integrated with wireless communication and solar charging features. The system collects behavioural and environmental data using various sensors while maintaining sustainable operation through renewable energy sources. Wireless connectivity allows remote monitoring [4].

J. O. Chelotti et al. presented a different automated sensing technologies and monitoring approaches used for livestock feeding behaviour analysis. The paper discusses accelerometers, computer vision systems, signal processing methods, and machine learning techniques used in precision livestock farming. The review highlights the importance of automated monitoring systems in improving feeding management, animal health, and farm productivity [5].

G. D. Siraskar et al. presented a low-cost automated cattle feeding system using affordable automation components and embedded technologies. The system was designed mainly for small and medium-scale farmers to reduce labor dependency and improve feeding consistency. The study emphasizes economical implementation while maintaining effective automation in livestock feeding operations [6].

Google Developers introduced TensorFlow Lite as a lightweight machine learning framework designed for embedded and edge AI applications. The framework supports deployment of AI models on low-power devices such as Raspberry Pi and microcontrollers. TensorFlow Lite is widely used for real-time AI inference in smart monitoring systems, including livestock behaviour analysis and automated decision-making applications [7].

M. Verma et al. presented an automated livestock feeding system controlled using embedded controllers. The feeding process was electronically scheduled and monitored to improve feeding accuracy and reduce manual intervention. The system demonstrates how embedded technologies can simplify livestock management and improve operational reliability in farms [8].

T. Ott et al. presented a monitoring system using accelerometer-based neck collars to track feeding duration and intake behaviour in dairy cows. The wearable sensor technology continuously collects motion data that helps identify feeding patterns and behavioural activities. The system provides accurate monitoring without requiring direct human observation [9].

L. Q. Mosquera et al. designed an automated cattle-feed dispensing system capable of delivering controlled quantities of feed at scheduled intervals. The system minimizes feed wastage and reduces human effort in feeding operations. The proposed solution demonstrates the effectiveness of automation in improving consistence [10].

2.1 Conclusion from Literature Survey

The literature survey shows that modern livestock management systems mainly use IoT, automation, computer vision and sensors to improve feeding efficiency and animal health monitoring. Several studies focused on automated feeding systems that use sensors, embedded controllers, and wireless communication to reduce manual labour, improve feeding accuracy, and minimize feed wastage. Other research highlighted the importance of computer vision and wearable sensors for continuous monitoring of cattle behaviour, and feeding duration, which helps improve farm productivity. Some studies emphasized sustainable and low-cost solutions by integrating solar energy and affordable automation methods suitable for small-scale farmers.

Precision feeding techniques based on historical production data were also found to enhance feed utilization and milk production efficiency.

Overall, the surveyed papers conclude that smart automation and monitoring technologies play a major role in improving efficiency, sustainability, and productivity in modern livestock farming.

Table 2.1 : Comparison Table of Existing Literature

SI. No.	Methods Used	Algorithm Used	Observation	Reference
1.	IoT, Sensors, Wireless Communication	IoT-based Automation	Automated livestock feeding and monitoring achieved	[1]
2.	Computer Vision, AI	Computer Vision Algorithms	Real-time cattle behavior monitoring possible	[2]
3.	Precision Feeding, Data Analysis	Historical Data Analysis	Improved feed utilization and milk production	[3]
4.	Multi-Sensors, Solar Power, Wireless Communication	Sensor Integration Techniques	Sustainable cattle monitoring system developed	[4]
5.	Automated Sensing, Signal Processing	Signal Processing Algorithms	Advanced livestock monitoring approaches reviewed	[5]
6.	Embedded Automation, Sensors	Embedded Control Algorithm	Low-cost automated feeding system implemented	[6]
7.	Edge AI, TensorFlow Lite	TensorFlow Lite Models	AI deployment on embedded systems enabled	[7]
8.	Embedded Controllers	Embedded Control System	Automated feeding control system developed	[8]

9.	Accelerometer Sensors, Wearable Devices	Accelerometer-based Monitoring	Feeding duration and intake monitored accurately	[9]
10.	Automated Feed Dispensing, Embedded Systems	Automated Dispensing Algorithm	Controlled cattle-feed dispensing achieved	[10]

2.2 Problem Statement

• Manual and Labor-Intensive Feeding Operations

Traditional cattle feeding methods require continuous human effort and regular monitoring. Farmers must manually feed cattle at scheduled intervals every day, making the process time-consuming and labor-intensive, especially in farms with a large number of cattle.

• Irregular Feeding and Feed Wastage

In many dairy farms, improper feeding schedules and inconsistent feed distribution negatively affect cattle health, growth, and milk production. Manual feeding methods may also lead to excessive feed wastage and inefficient feed utilization.

• Lack of Intelligent Cattle Detection

Most conventional automated feeding systems operate using fixed timers or predefined schedules without verifying the actual presence of cattle. This reduces feeding accuracy and may result in unnecessary feed dispensing when cattle are not present.

• Difficulty in Continuous Livestock Monitoring

Continuous monitoring of cattle inside large farm environments is difficult for farmers and requires significant manpower. Lack of intelligent monitoring systems may reduce operational efficiency and increase dependency on manual supervision.

2.3 Solution to the Problem

- **Automated Cattle Feeding Operation**

The proposed automatic cattle feeding bot automates the cattle feeding process using Raspberry Pi, DC motors, and servo-controlled feed dispensing mechanism. The robot autonomously moves inside the farm environment and dispenses feed automatically without requiring continuous human intervention.

- **AI-Based Real-Time Cattle Detection**

The system uses a USB web camera along with OpenCV and TensorFlow Lite AI model for real-time cattle detection. The artificial intelligence model continuously analyzes captured image frames and confirms cattle presence before activating the feed dispensing mechanism.

- **Controlled Feed Dispensing Mechanism**

A servo motor-controlled feed hopper is used to dispense feed in controlled quantity only when cattle are detected successfully. This minimizes feed wastage and improves feeding consistency in dairy farms.

- **Continuous Autonomous Monitoring**

The robot continuously monitors the surrounding farm environment using camera-based image acquisition and autonomous movement control. This reduces human supervision requirements and improves operational efficiency in livestock management applications.

Chapter 3

METHODOLOGY

The methodology adopted for the automatic cattle feeding bot focuses on developing an intelligent and autonomous livestock feeding system using computer vision, embedded systems, and robotic automation.

The proposed system is designed to automate the cattle feeding process by detecting cattle in real time and dispensing feed automatically without requiring continuous human supervision. The methodology mainly involves image acquisition, camera based cattle detection, robot movement control, and automatic feed dispensing using a servo-controlled mechanism.

The complete architecture of the proposed system is shown in fig. 3.1. The system mainly consists of a Raspberry Pi 5, camera module, motor driver, DC motors, feed dispensing mechanism, and power supply unit. The Raspberry Pi acts as the central processing and control unit of the entire system.

The camera module continuously captures real-time images from the surroundings and sends them to the Raspberry Pi for processing. The captured images are processed using OpenCV and TensorFlow Lite AI model for cattle detection. Based on the detection result, the Raspberry Pi controls the movement of the robot and activates the feed dispensing mechanism automatically.

3.1 Block Diagram

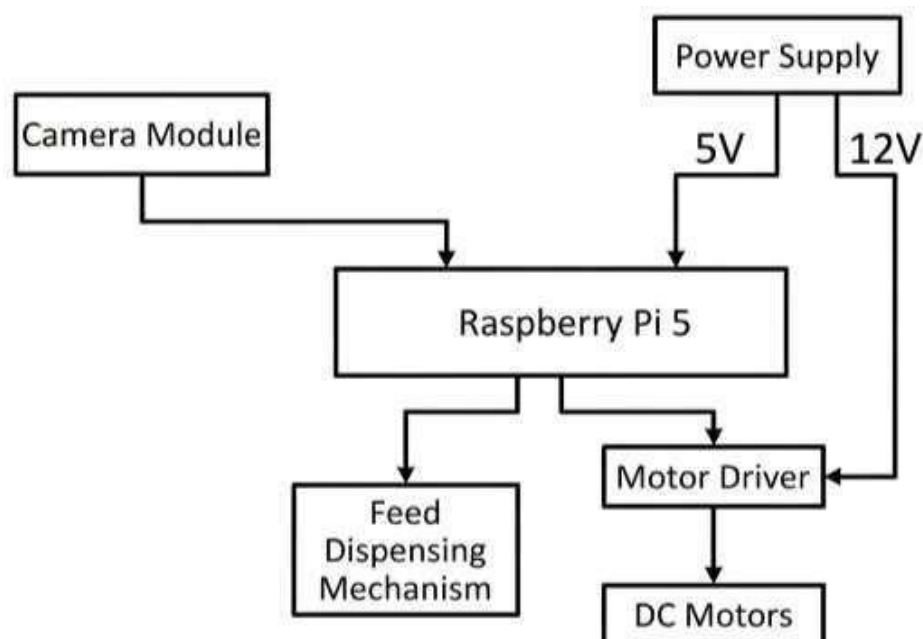


Fig. 3.1 : Block Diagram of Proposed System

The block diagram of the automatic cattle feeding bot is shown in fig. 3.1. The proposed system is designed using an integrated architecture where all hardware and software components work together to perform autonomous cattle feeding operations. The system mainly includes a Raspberry Pi 5, camera module, motor driver, DC motors, feed dispensing mechanism, and power supply section. The Raspberry Pi serves as the main controller and coordinates all operations of the system.

The camera module is connected directly to the Raspberry Pi and continuously captures real-time video frames from the surrounding environment. These captured images are processed using OpenCV image processing techniques and TensorFlow Lite AI model to detect the presence of cattle.

The AI model compares the captured image features with trained cattle datasets and identifies cattle based on confidence levels. If cattle are not detected, the robot continues normal movement inside the farm environment. When cattle are detected successfully, the Raspberry Pi sends control signals to stop the robot movement and activate the feed dispensing mechanism.

The motor driver shown in fig. 3.1 acts as an interface between the Raspberry Pi and DC motors. Since the Raspberry Pi GPIO pins cannot provide sufficient current to drive motors directly, the motor driver amplifies the control signals and supplies the required voltage and current to the motors.

The DC motors are responsible for moving the robot autonomously inside the farm. During normal operation, the motors rotate continuously and move the robot forward. When cattle are detected, the Raspberry Pi immediately stops the motors through the motor driver.

The feed dispensing mechanism is controlled using a servo motor connected to the feed hopper. After cattle detection, the Raspberry Pi activates the servo motor, which rotates and opens the feed outlet flap. Feed is dispensed automatically in controlled quantity for a fixed duration. After dispensing is completed, the servo motor returns to its original position and closes the feed outlet flap automatically. The robot then resumes autonomous movement and continues monitoring for cattle.

The power supply arrangement shown in fig. 3.1 consists of two voltage levels. A regulated 5V supply is used to power the Raspberry Pi and servo motor, while a 12V supply is provided to the motor driver and DC motors. This dual-power arrangement ensures stable operation of both low-power electronic components and high-power mechanical components used in the system.

The working methodology of the Automatic Cattle Feeding Bot begins with initializing all hardware modules connected to the Raspberry Pi. Once the system is powered ON, the Raspberry Pi activates the camera module, motor driver, DC motors, and feed dispensing mechanism. The

robot begins autonomous movement inside the farm environment using the DC motors controlled through the motor driver.

While the robot moves forward, the camera module continuously captures real-time video frames from the surroundings. These captured frames are transferred to the Raspberry Pi for image processing.

OpenCV is used for preprocessing operations such as resizing, normalization, and feature extraction from the captured images. The processed images are then analyzed using the TensorFlow Lite AI model trained for cattle detection.

The model identifies cattle by comparing extracted image features with trained datasets stored in the model. If the confidence level exceeds the predefined threshold value, cattle detection is confirmed successfully.

During normal operation, when cattle are not detected, the robot continues autonomous movement without interruption. When cattle are detected, the Raspberry Pi immediately sends control signals to the motor driver to stop the DC motors. After stopping the robot movement, the Raspberry Pi activates the servo motor connected to the feed dispensing mechanism.

The servo motor rotates from its initial position and opens the flap door of the feed hopper. Feed falls through the outlet and is dispensed automatically in controlled quantity. After dispensing feed for a fixed duration, the servo motor rotates back to its original position and closes the feed outlet flap automatically.

Once the feed dispensing process is completed, the Raspberry Pi restarts the DC motors through the motor driver, and the robot resumes autonomous movement inside the farm environment. This entire process operates continuously without human intervention and ensures timely and accurate cattle feeding operations.

3.2 Hardware Requirements

3.2.1 Raspberry Pi 5



Fig. 3.2: Raspberry Pi 5 [11]

The Raspberry Pi 5 serves as the main processing and control unit of the automatic cattle feeding bot. The image of the Raspberry Pi 5 is shown in fig. 3.2. It is a compact and high-performance single-board computer developed for embedded systems, robotics, and IoT applications. The Raspberry Pi 5 is equipped with a powerful processor, GPIO pins, USB interfaces, HDMI support, camera interface, and wireless connectivity features, making it highly suitable for autonomous robotic systems.

In the proposed system, the Raspberry Pi performs all major operations including image acquisition, image processing, cattle detection, robot movement control, and feed dispensing control. The Raspberry Pi receives real-time video frames from the camera module and processes the captured images using OpenCV image processing techniques. The TensorFlow Lite AI model executes directly on the Raspberry Pi and performs real-time cattle detection by comparing image features with trained datasets.

The Raspberry Pi also controls the motor driver through GPIO pins and manages the movement of the DC motors according to the cattle detection result. When cattle are detected successfully, the Raspberry Pi sends control signals to stop the robot movement and activate the servo motor connected to the feed dispensing mechanism. The entire control program is implemented using Python programming language due to its simplicity, flexibility, and support for image processing libraries.

The Raspberry Pi 5 offers several advantages such as high processing capability, low power consumption, compact size, and easy interfacing. The use of Raspberry Pi makes the system cost-effective, portable, and suitable for modern smart farming applications.

3.2.3 USB Web Camera Module



Fig. 3.3: USB Web Camera Module [12]

The USB web camera module is used for real-time image acquisition and cattle detection in the automatic cattle feeding bot. The image of the USB web camera module is shown in fig. 3.3. The web camera is connected directly to the Raspberry Pi through the USB interface and continuously captures video frames from the surrounding environment while the robot moves inside the farm.

The web camera plays an important role in enabling computer vision operations in the proposed system. During operation, the camera continuously scans the surroundings and captures images of cattle present near the robot. These captured video frames are transferred to the Raspberry Pi for processing using OpenCV and TensorFlow Lite libraries.

OpenCV is used for image preprocessing operations such as image resizing, normalization, filtering, and feature extraction. The processed frames are then analyzed using the TensorFlow Lite AI model trained for cattle detection. The artificial intelligence model identifies cattle based on trained image datasets and confidence threshold values. If cattle are detected successfully, the Raspberry Pi initiates the feed dispensing process automatically.

The use of a camera-based detection system improves feeding accuracy and reduces feed wastage because feed is dispensed only when cattle are present. The USB web camera also enables continuous real-time monitoring and intelligent autonomous operation of the feeding bot.

The USB web camera is selected because of its simple interfacing, good image quality, low cost, and compatibility with Raspberry Pi systems. It supports real-time video streaming and provides reliable performance for smart farming applications.

3.2.4 Motor Driver

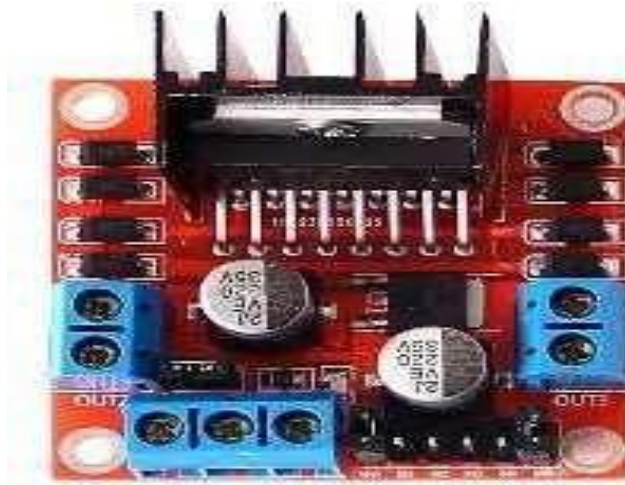


Fig. 3.4: Motor Driver [13]

The motor driver acts as an interface between the Raspberry Pi and the DC motors used for robot movement. The image of the motor driver is shown in fig. 3.4. Since the GPIO pins of the Raspberry Pi cannot provide sufficient current and voltage required to drive motors directly, the motor driver amplifies the control signals and supplies the necessary power to the motors.

In the proposed system, an L298N motor driver module is used to control the movement and direction of the DC motors. The motor driver receives control signals from the Raspberry Pi and controls the operation of the motors accordingly. During normal operation, the Raspberry Pi sends signals to move the robot forward continuously while monitoring for cattle detection.

When cattle are detected successfully, the Raspberry Pi immediately sends stop signals to the motor driver. The motor driver then stops the motors, causing the robot to stop near the cattle for feed dispensing. After dispensing is completed, the Raspberry Pi again sends control signals to restart the motors and continue robot movement.

The motor driver provides safe and efficient motor operation while protecting the Raspberry Pi from excessive current flow and voltage fluctuations. It ensures stable operation of the motors and improves the reliability of the robotic system. The use of the L298N motor driver also simplifies interfacing and allows smooth control of robot movement in autonomous farming applications.

3.2.5 DC Motor



Fig. 3.5: DC Motor[14]

DC gear motors are used for autonomous movement of the automatic cattle feeding bot. The image of the DC motor is shown in fig.3.5. These motors convert electrical energy into mechanical motion and generate sufficient torque required to move the robot carrying the feed hopper and hardware components.

The DC motors are connected to the motor driver and controlled by the Raspberry Pi. During normal operation, the motors rotate continuously and move the robot forward inside the farm environment. The robot autonomously travels while the camera captures images for cattle detection. The motor driver controls the speed and direction of the motors based on signals received from the Raspberry Pi.

When cattle are detected successfully, the Raspberry Pi sends stop commands to the motor driver, and the motors stop immediately. This allows accurate positioning of the robot near the cattle before feed dispensing begins. After feed dispensing is completed, the motors restart automatically and continue robot movement.

DC gear motors are selected because they provide high torque, reliable operation, simple control, and low cost. The gear mechanism attached to the motors reduces speed and increases torque, which is necessary for moving the robot smoothly on different surfaces inside the farm environment. The motors play a crucial role in enabling autonomous robotic movement and intelligent feeding operations.

3.2.6 Servo Motor Based Feed Dispensing Mechanism



Fig. 3.6: Servo Motor Based Feed Dispensing Mechanism [15]

The servo motor is used to control the feed dispensing mechanism of the automatic cattle feeding bot. The image of the servo motor is shown in fig.3.6. A feed hopper with a flap door arrangement is connected to the servo motor to enable automatic dispensing of feed.

The servo motor provides precise angular movement and operates based on PWM control signals generated by the Raspberry Pi. When cattle are detected successfully, the Raspberry Pi activates the servo motor through GPIO pins. The servo motor rotates from its initial position and opens the flap door of the feed hopper.

As the flap door opens, feed falls through the outlet and is dispensed automatically in controlled quantity. After dispensing feed for a fixed duration, the Raspberry Pi sends signals to rotate the servo motor back to its original position. The flap door closes automatically, stopping the feed flow completely.

The use of a servo-controlled dispensing mechanism ensures accurate and controlled feed delivery with minimum wastage. Servo motors are highly suitable for this application because they provide precise position control, fast response, low power consumption, and reliable operation. The feed dispensing mechanism enables fully automatic feeding without human intervention and improves feeding consistency in dairy farms.

3.2.7 Power Supply



Fig. 3.7: Power Supply [16]

The power supply unit provides electrical energy to all hardware components used in the automatic cattle feeding bot. The image of the power supply arrangement is shown in fig.3.7. Proper power management is essential for ensuring stable and uninterrupted operation of the robotic system.

The proposed system uses a dual-power arrangement consisting of a regulated 5V supply and a 12V supply. The Raspberry Pi 5, camera module, and servo motor operate using a regulated 5V power supply. Since these electronic components require stable low-voltage operation, a regulated power source is necessary to avoid voltage fluctuations and system instability.

The DC motors require higher voltage and current to generate sufficient torque for robot movement. Therefore, a 12V supply is provided to the motor driver and DC motors. The motor driver distributes the required current and voltage to the motors according to movement control signals received from the Raspberry Pi.

The dual-power arrangement ensures proper operation of both low-power electronic components and high-power mechanical components used in the system. The power supply unit enables reliable operation of the robot during autonomous movement, image processing, cattle detection, and feed dispensing operations. Proper power management also improves system stability, efficiency, and overall performance of the automatic cattle feeding bot.

3.3 Software Requirements

3.3.1 Raspberry Pi OS

Raspberry Pi OS is the operating system used in the automatic cattle feeding bot for controlling all software and hardware operations. Raspberry Pi OS is a linux based operating system specially developed for Raspberry Pi boards and embedded applications. It provides a stable and efficient platform for running python programs, image processing libraries, and hardware interfacing modules.

In the proposed system, Raspberry Pi OS manages the execution of OpenCV, TensorFlow Lite, GPIO control libraries, and Python scripts used for autonomous robot operation. The operating system allows the Raspberry Pi to interface with the USB web camera, motor driver, DC motors, and servo motor efficiently.

Raspberry Pi OS also supports multitasking and real-time execution of programs required for image acquisition, cattle detection, motor control, and feed dispensing operations. The operating system provides a user-friendly environment, simple package management, and support for IoT applications, making it highly suitable for smart farming systems.

3.3.2 Python Programming Environment

Python is the primary programming language used for implementing the automatic cattle feeding bot. Python is widely used in embedded systems, robotics, and image processing applications because of its simplicity, flexibility, and extensive library support.

In the proposed system, python is used to write the control program for image processing, cattle detection, motor control, and feed dispensing operations. Python libraries such as OpenCV, TensorFlow Lite, GPIO libraries, and NumPy are used for implementing different functionalities of the system.

The python program continuously captures video frames from the USB web camera, processes the images using OpenCV, and executes the TensorFlow Lite AI model for cattle detection. Based on the detection result, the program controls the movement of the robot and activates the servo motor for feed dispensing.

Python provides simple syntax, easy debugging, fast development, and excellent compatibility with Raspberry Pi systems. These features make python highly suitable for implementing intelligent autonomous robotic systems and smart farming applications.

3.3.3 OpenCV

OpenCV (Open source Computer Vision Library) is used for image processing and computer vision operations in the automatic cattle feeding bot. OpenCV provides various functions and algorithms required for processing images and analyzing video frames captured from the USB web camera.

In the proposed system, OpenCV is used for image preprocessing operations such as image resizing, normalization, filtering, and feature extraction. The captured video frames are processed using OpenCV before being analyzed by the TensorFlow Lite AI model for cattle detection.

OpenCV enables real-time image processing and improves the efficiency of cattle detection operations. It allows the Raspberry Pi to process continuous video frames and identify cattle accurately under different environmental conditions.

The OpenCV library is widely used in robotics, and computer vision applications because of its fast processing speed, extensive functionality, and compatibility with python and Raspberry Pi systems. The use of OpenCV improves the intelligence and automation capability of the proposed feeding bot.

3.3.4 TensorFlow Lite

TensorFlow Lite is used for implementing the artificial intelligence based cattle detection model in the automatic cattle feeding bot. TensorFlow Lite is a lightweight machine learning framework developed for embedded systems and edge devices such as Raspberry Pi.

In the proposed system, the TensorFlow Lite model is trained using cattle image datasets and used for real-time cattle detection. The processed image frames received from OpenCV are analyzed by the TensorFlow Lite AI model to identify cattle based on trained image features and confidence levels.

When cattle are detected successfully, the TensorFlow Lite model sends the detection result to the Python control program running on the Raspberry Pi. Based on this result, the Raspberry Pi controls the motor driver and feed dispensing mechanism automatically.

TensorFlow Lite is selected because of its fast inference speed, low memory usage, and compatibility with embedded systems. It enables real-time execution on the Raspberry Pi and improves the intelligence and automation capability of the proposed system. The use of TensorFlow Lite makes the feeding bot efficient, intelligent, and suitable for smart farming applications.

Chapter 4

SYSTEM DESIGN

The system design of the automatic cattle feeding bot integrates Raspberry Pi 5, camera module, motor driver, and servo-based feed dispensing unit for autonomous operation. It enables real-time cattle detection, movement control, and automated feed delivery using camera based image processing and embedded control.

4.1 Circuit Diagram

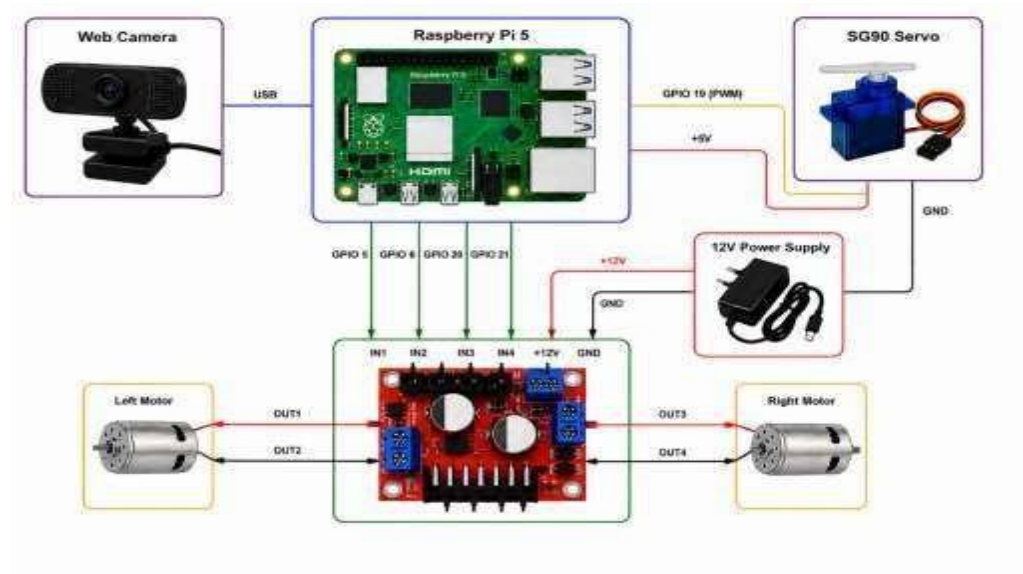


Fig. 4.1: Circuit Diagram of the Proposed System

The circuit diagram of the automatic cattle feeding bot is shown in fig. 4.1. The system mainly consists of a Raspberry Pi 5, USB web camera, L298N motor driver, DC motors, servo motor, and power supply unit. The Raspberry Pi acts as the central controller and coordinates all operations of the system.

The USB web camera is connected directly to the Raspberry Pi through the USB port. The camera continuously captures real-time video frames from the surroundings and transfers them to the Raspberry Pi for cattle detection and image processing operations.

The L298N motor driver is connected to the GPIO pins of the Raspberry Pi for controlling the movement of the DC motors. GPIO pins 5 and 6 are connected to the left motor control inputs, while GPIO pins 20 and 21 are connected to the right motor control inputs. The motor driver controls the direction and movement of the robot based on signals received from the Raspberry Pi.

The servo motor used for feed dispensing is connected to GPIO pin 19 of the Raspberry Pi. The servo motor controls the opening and closing of the feed dispensing flap connected to the feed hopper. When cattle are detected successfully, the Raspberry Pi activates the servo motor to dispense feed automatically. The DC motors are connected to the output terminals of the L298N motor driver and are responsible for autonomous movement of the robot inside the farm environment. The motor driver receives power from a 12V supply to provide sufficient torque for robot movement.

The Raspberry Pi, USB web camera, and servo motor operate using a regulated 5V power supply. The dual-power arrangement ensures stable operation of both low-power electronic components and high-power mechanical components used in the system. The circuit diagram enables efficient integration of image processing, motor control, and autonomous feed dispensing operations required for the automatic cattle feeding bot.

4.2 Flow Diagram

The flowchart illustrates the step-by-step working process of the automatic cattle feeding bot system. It shows how the system detects cattle, processes the input, and automatically dispenses feed in a controlled manner.

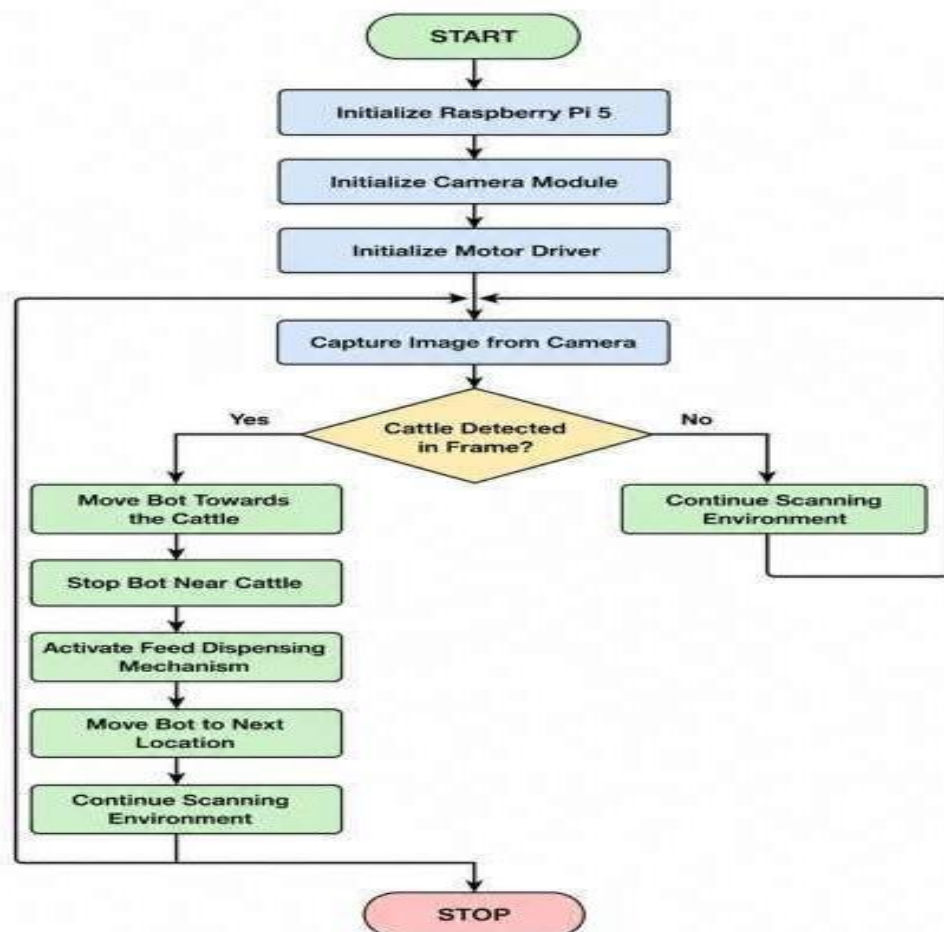


Fig. 4.2: Flow Diagram of Automatic Cattle Feeding Bot

The flow diagram of the automatic cattle feeding bot is shown in fig.4.2. The flowchart represents the overall operational sequence and working process of the proposed autonomous feeding system. The system begins by initializing the Raspberry Pi 5, camera module, and motor driver required for image processing, cattle detection, robot movement, and feed dispensing operations.

After initialization, the USB web camera continuously captures real-time video frames from the surrounding environment. These captured images are transferred to the Raspberry Pi for processing using OpenCV and TensorFlow Lite AI model. The artificial intelligence model analyzes the captured frames and checks whether cattle are present in the frame.

If cattle are not detected, the system continues scanning the environment continuously by capturing additional image frames. This process repeats automatically until cattle are identified successfully. The continuous scanning operation enables real-time monitoring and intelligent autonomous functioning of the feeding bot.

When cattle are detected successfully, the robot moves towards the cattle automatically using DC motors controlled through the motor driver. Once the robot reaches near the cattle, the Raspberry Pi stops the robot movement and activates the feed dispensing mechanism.

The feed dispensing mechanism consists of a servo-controlled flap connected to the feed hopper. The servo motor rotates and opens the flap door, allowing feed to fall through the outlet automatically in controlled quantity.

After feed dispensing is completed, the flap closes automatically and the robot moves towards the next location for further monitoring.

The system then resumes continuous scanning of the surrounding environment for cattle detection. The complete process operates in a continuous loop, enabling autonomous cattle monitoring and automatic feed dispensing without requiring human intervention. The proposed flow ensures timely feeding, reduced manual labor, minimized feed wastage, and efficient smart farming operation.

4.3 Pin Diagram

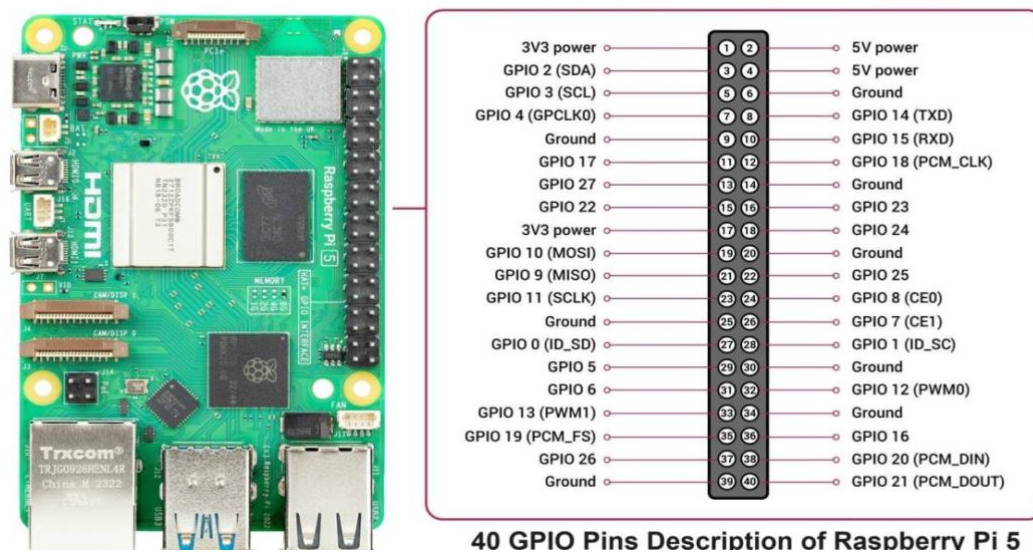


Fig.4.3: Raspberry Pi 5 Pin Diagram [17]

The Raspberry Pi 5 is a powerful single-board computer widely used in embedded systems, robotics, and automation applications. One of its major features is that as shown in fig.4.3 there are 40 General Purpose Input/Output (GPIO) pins, which enable communication between the Raspberry Pi and external electronic devices such as sensors, motors, cameras, relays, and displays.

The GPIO header contains 40 pins arranged in two rows, where each pin performs a specific function including power supply, grounding, digital input/output operations, and communication protocols such as UART, SPI, I2C, and PWM. The 5V and 3.3V pins provide electrical power to connected modules, while multiple ground pins complete the electrical circuit and ensure stable operation of components.

The GPIO pins can be programmed either as input pins to receive signals from sensors or as output pins to control actuators such as LEDs, relays, and motors. Communication protocols supported by these pins improve the interfacing capability of the Raspberry Pi. The I2C protocol uses GPIO 2 and GPIO 3 for connecting sensors and display modules through two-wire communication.

UART communication, available through GPIO 14 and GPIO 15, enables serial data transmission with external devices such as microcontrollers and GPS modules. SPI communication uses GPIO 7, 8, 9, 10, and 11 for high-speed data transfer with devices like displays and memory modules. PWM-supported GPIO pins such as GPIO 12, GPIO 13, GPIO 18, and GPIO 19 are used for controlling motor speed, servo motor rotation, and LED brightness.

In the cattle feeding robot system, the Raspberry Pi 5 GPIO pins play a significant role in interfacing and controlling different hardware components. The GPIO pins communicate with the L298N Motor Driver to control the movement and direction of DC motors used for robot navigation.

PWM-enabled GPIO pins are used to operate the SG90 Servo Motor that controls the feed dispensing mechanism. Additionally, the Raspberry Pi processes images captured from the USB camera module and executes the TensorFlow Lite AI model for real-time cattle detection. Therefore, the GPIO pins act as the central communication interface that enables intelligent automation, motor control, and real-time interaction between the software and hardware components of the robotic system.

4.4 Software Implementation

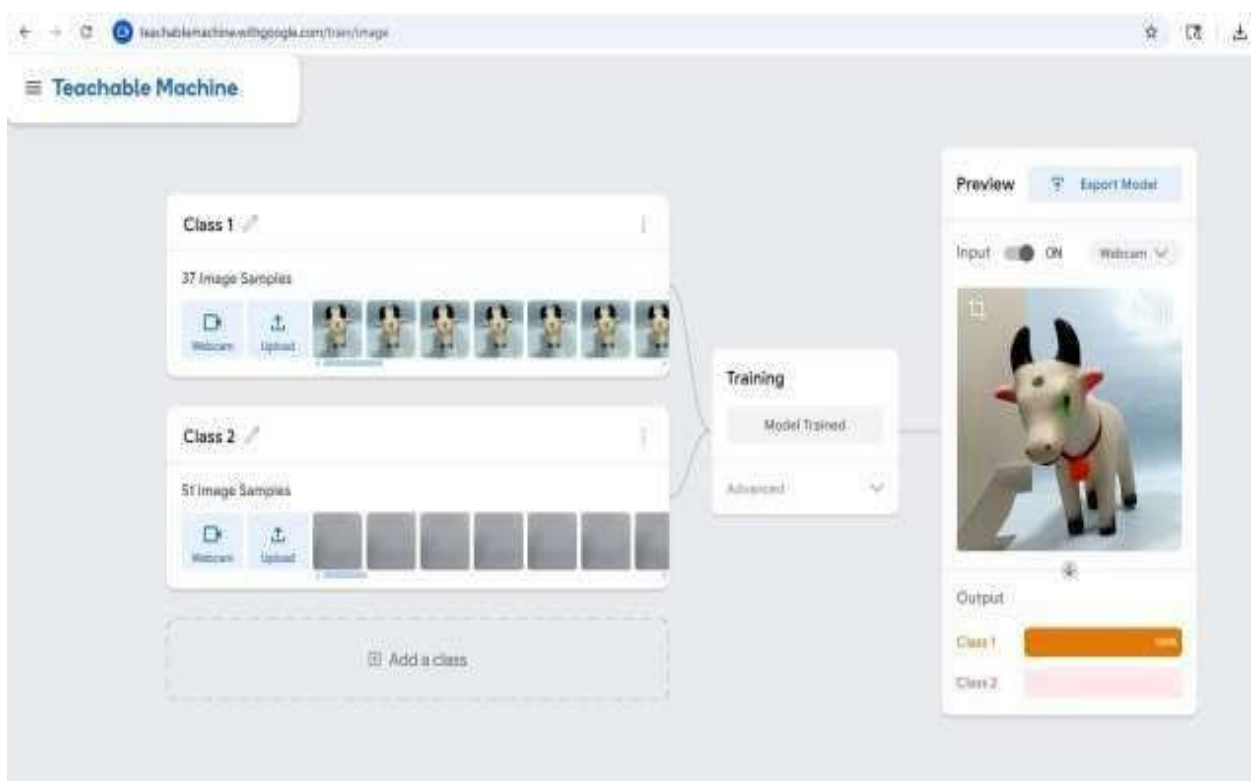


Fig. 4.4: TensorFlow Lite Model Training Using Teachable Machine

The software implementation of the automatic cattle feeding bot is developed using python programming language on the Raspberry Pi 5 platform. The software controls image acquisition, cattle detection, robot movement, feed dispensing, and autonomous navigation operations. OpenCV, TensorFlow Lite, NumPy, and time libraries are used for implementing different functionalities of the system. Initially, the required python libraries are imported into the program. The OpenCV library is used for image acquisition and image processing operations, while NumPy is used for numerical processing and array handling.

The gpiozero library is used for controlling the DC motors and servo motor through Raspberry Pi GPIO pins. TensorFlow Lite runtime is used for loading and executing the trained artificial intelligence model for cattle detection.

The TensorFlow Lite cattle detection model used in the system is developed using Google Teachable Machine platform, as shown in fig. 4.4. Teachable machine is a web-based machine learning platform that allows users to train image classification models without complex programming. In the proposed system, two different classes are created for training the model. The first class contains cattle images, while the second class contains background or non-cattle images.

Multiple image samples are collected using the webcam and uploaded into the teachable machine platform. During the training process, the model automatically extracts image features and learns the visual characteristics of cattle from the dataset. After successful training, the model is tested using real-time webcam input to verify accurate detection performance. The trained model is then exported in TensorFlow Lite format and stored as model in the Raspberry Pi for real-time inference operations.

The software initializes the left and right DC motors using the gpiozero motor class. GPIO pin 5 is connected to IN1 and GPIO pin 6 is connected to IN2 of the L298N motor driver for controlling the left motor. Similarly, GPIO pin 20 is connected to IN3 and GPIO pin 21 is connected to IN4 for controlling the right motor. Separate motor control functions such as forward, reverse, and stop are implemented in the program to control the autonomous movement of the robot.

The SG90 servo motor used for feed dispensing is connected to GPIO pin 19 of the Raspberry Pi. The servo motor is controlled using PWM signals generated by the Raspberry Pi through the gpiozero servo library. A dedicated servo control function named servo action is implemented in the program. When cattle are detected successfully, the robot stops automatically and the servo motor rotates to open the feed dispensing flap. Feed is dispensed for a fixed duration, after which the servo motor returns to its original position and closes the flap.

The USB web camera connected to the Raspberry Pi continuously captures real-time video frames from the surrounding environment using OpenCV. The captured image frames are resized to 224×224 resolution before being processed by the TensorFlow Lite model. Image normalization is also performed to improve model accuracy and inference performance.

The TensorFlow Lite interpreter loads the trained artificial intelligence model and performs inference on the captured image frames. The model analyzes the input image and predicts whether cattle are present in the frame. The output prediction contains confidence values corresponding to different image classes. The program calculates the confidence value using NumPy functions and compares it with the predefined threshold value of 0.90.

When the confidence value exceeds the threshold limit and cattle are detected successfully, the Raspberry Pi stops the robot movement immediately and activates the feed dispensing mechanism. A cooldown mechanism is also implemented in the software to avoid repeated detection of the same cattle within a short duration. This improves feeding accuracy and prevents unnecessary repeated feed dispensing operations.

The software is designed to feed a maximum of four cattle during one operation cycle. A counter variable is used to monitor the number of cattle detected and fed successfully. After feeding four cattle, the Raspberry Pi calculates the total forward movement time of the robot. The robot then moves backward for the same duration using reverse motor control, allowing it to return approximately to its starting position automatically.

The complete software operates inside a continuous while loop, enabling uninterrupted environment scanning, real-time cattle detection, autonomous robot movement, and automatic feed dispensing operations. OpenCV display functions are also used for monitoring real-time camera output during system operation.

The implemented software enables efficient integration of image processing, motor control, and embedded automation technologies for smart livestock feeding applications. The software provides autonomous operation, reduced manual labor, real-time cattle monitoring, and accurate feed dispensing functionality for modern smart farming systems.

4.5 Hardware Implementation

The hardware implementation of the automatic cattle feeding bot involves the integration of Raspberry Pi 5, USB web camera, L298N motor driver, DC motors, servo motor, and power supply unit to perform autonomous cattle detection and automatic feed dispensing operations. The complete hardware arrangement is shown in fig.4.1. The Raspberry Pi 5 acts as the central controller of the entire system and coordinates all hardware operations.

The USB web camera is connected directly to the Raspberry Pi through the USB interface and continuously captures real-time video frames from the surrounding environment. The captured images are processed by the Raspberry Pi for cattle detection using OpenCV and TensorFlow Lite AI model.

The L298N motor driver is connected to the GPIO pins of the Raspberry Pi for controlling the movement of the DC motors. GPIO pin 5 is connected to IN1, GPIO pin 6 is connected to IN2, GPIO pin 20 is connected to IN3, and GPIO pin 21 is connected to IN4 of the L298N motor driver. These GPIO connections allow the Raspberry Pi to control the direction and movement of the left and right DC motors.

Two DC gear motors are connected to the output terminals of the L298N motor driver. The left motor is connected to OUT1 and OUT2, while the right motor is connected to OUT3 and OUT4 of the motor driver. During normal operation, the motors rotate and move the robot autonomously inside the farm environment. The motor driver receives control signals from the Raspberry Pi and drives the motors accordingly.

The SG90 servo motor is connected to GPIO pin 19 of the Raspberry Pi for controlling the feed dispensing mechanism. The servo motor operates using PWM control signals generated by the Raspberry Pi. A feed hopper with a flap door arrangement is connected to the servo motor. When cattle are detected successfully, the servo motor rotates and opens the feed outlet flap, allowing feed to fall automatically. After dispensing feed, the servo motor returns to its original position and closes the flap.

The entire system is powered using a 12V power supply connected to the L298N motor driver. The motor driver distributes the required voltage and current to the DC motors for robot movement. Proper grounding connections are maintained between all hardware modules to ensure stable operation of the system.

Chapter 5

RESULTS AND DISCUSSION

The automatic cattle feeding bot was successfully designed and implemented using Raspberry Pi 5, TensorFlow Lite AI model, OpenCV image processing, USB web camera, L298N motor driver, DC motors, and SG90 servo motor. The developed system was tested under different operating conditions to verify the performance of cattle detection, robot movement, and automatic feed dispensing operations.

The USB web camera successfully captured real-time video frames from the surrounding environment and transferred the images to the Raspberry Pi for processing. OpenCV image processing techniques and TensorFlow Lite AI model were able to analyze the captured frames efficiently and perform real-time cattle detection operations.

The tensorflow lite model trained using google teachable machine successfully identified cattle images with good accuracy under normal lighting conditions. The model detected cattle whenever the confidence value exceeded the predefined threshold value of 0.90. The trained artificial intelligence model responded quickly and enabled real-time detection during robot movement operations.

The autonomous movement of the robot was achieved successfully using DC motors controlled through the L298N motor driver. During normal operation, the robot moved continuously inside the testing environment while scanning for cattle presence using the USB web camera. The movement control functions implemented in the software enabled smooth forward movement, stopping, and reverse movement operations.

When cattle were detected successfully, the Raspberry Pi immediately stopped the robot and activated the SG90 servo motor connected to the feed dispensing mechanism. The servo motor successfully opened the flap door of the feed hopper and dispensed feed automatically in controlled quantity. After feed dispensing was completed, the flap closed automatically and the robot resumed movement for further cattle monitoring.

The cooldown mechanism implemented in the software successfully prevented repeated detection of the same cattle within a short duration. This improved feeding accuracy and reduced unnecessary feed dispensing operations. The system also successfully monitored the number of cattle fed using the counter variable implemented in the software. The return mechanism implemented in the software operated successfully after feeding four cattle. The Raspberry Pi calculated the total forward movement time and moved the robot backward for the same duration, allowing the robot to return approximately to its starting position automatically.

The proposed system reduced manual labor involved in livestock feeding operations and improved automation in smart farming applications. The integration of image processing, embedded systems, and robotic automation enabled intelligent cattle monitoring and automatic feed dispensing functionality. The developed automatic cattle feeding bot demonstrated reliable operation, real-time cattle detection, autonomous movement, and controlled feed dispensing performance. The system can be further improved in future by integrating obstacle detection sensors, GPS tracking, and wireless monitoring.

Result Image :

The image shows a prototype model of an automatic cattle feeding bot system designed for smart farming applications. The setup includes a miniature cattle shed, feeding robot cart, road path, and grass field representation.

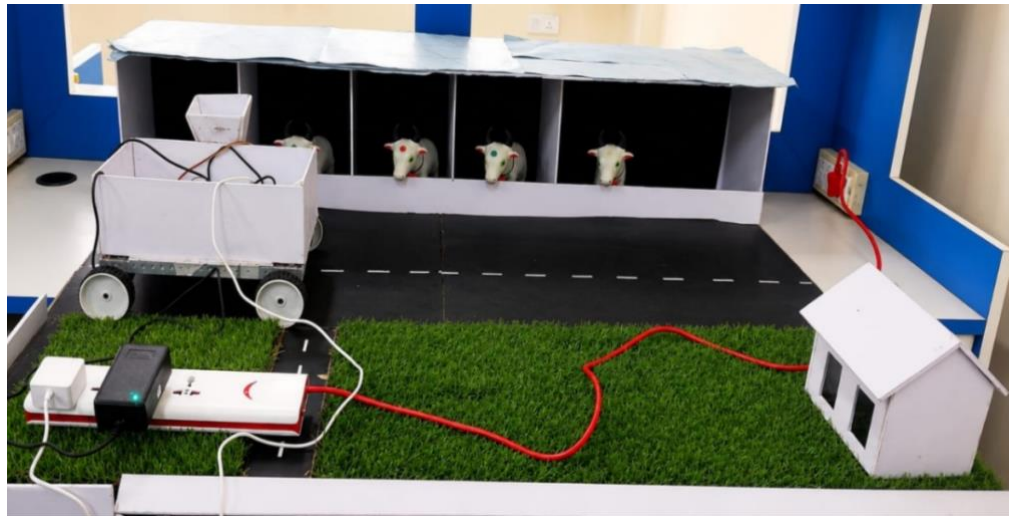


Fig. 5.1: Design Model of Automatic Cattle Feeding Bot

5.1 Advantages

The following are some advantages of the automatic cattle feeding bot

- **Reduces Manual Labour**

The robot automatically dispenses feed, reducing the need for farmers to manually feed cattle.

- **Autonomous Operation**

Uses camera-based detection to identify cattle and perform feeding automatically.

- **Reduced Human Supervision**

The automated feeding system reduces the need for constant human monitoring.

- **Smart Monitoring Features**

Camera and sensor integration can help monitor cattle presence and feeding activity.

- **Reduces Feed Wastage**

Controlled dispensing minimizes unnecessary spillage and wastage of feed materials.

5.2 Limitations

The following are some limitations of the automatic cattle feeding bot

- **Limited Feed Capacity**

The small hopper size limits the amount of feed the robot can carry, requiring frequent refilling.

- **Dependence on Camera Detection**

The system relies on camera detection, so poor lighting or obstruction may affect cattle detection accuracy.

5.3 Applications

The following are some applications of the automatic cattle feeding bot

- **Dairy Farms**

The automatic cattle feeding bot can be used in small and medium dairy farms for automatic cattle feeding operations. The system reduces manual effort involved in feeding cattle and ensures feed is dispensed in controlled quantities at regular intervals. This improves feeding efficiency and helps farmers manage livestock more effectively.

- **Smart Livestock Farming**

The proposed system can be used as a part of modern smart livestock farming and precision agriculture systems. The integration of image processing, and autonomous robotic movement enables intelligent cattle monitoring and automated feeding operations. The system supports advanced smart farming technologies for improving livestock management and farm productivity.

- **Agricultural Automation Research**

The automatic cattle feeding bot can be used as a research prototype for robotics and automation applications in agriculture. The system demonstrates the implementation of TensorFlow Lite, OpenCV, Raspberry Pi, and embedded systems for autonomous farming applications. It can be further enhanced for future agricultural automation and smart farming research projects.

Chapter 6

CONCLUSION

The project titled “Automatic Cattle Feeding Bot” was successfully designed and implemented to automate cattle feeding operations using image processing, and embedded system technologies. The proposed system was developed to reduce manual labor involved in livestock feeding and improve efficiency in smart farming applications. The system uses a Raspberry Pi 5 as the central controller for performing image processing, cattle detection, robot movement control, and automatic feed dispensing operations.

The automatic cattle feeding bot integrates major hardware components such as a USB web camera, L298N motor driver, DC motors, SG90 servo motor, and power supply unit. The USB web camera continuously captures real-time video frames from the surrounding environment, while OpenCV and TensorFlow lite AI model are used for cattle detection operations. The trained TensorFlow Lite model developed using google teachable machine successfully identified cattle images and enabled intelligent decision-making during autonomous robot operation.

The robot was capable of moving autonomously inside the testing environment using DC motors controlled through the L298N motor driver. When cattle were detected successfully, the robot stopped automatically and activated the servo-controlled feed dispensing mechanism.

The implemented software enabled real-time cattle detection, autonomous navigation, feed dispensing, cooldown operation, and automatic return movement after feeding operations. The project demonstrated successful integration of computer vision, embedded systems, and robotic automation for smart livestock management applications.

Overall, the automatic cattle feeding bot provides a cost-effective, portable, and intelligent solution for modern dairy farms and smart farming applications. The proposed system reduces manual labor, improves feeding consistency, minimizes feed wastage, and demonstrates the potential of camera based automation technologies in livestock management systems.

6.1 Future Scope

The automatic cattle feeding bot prototype demonstrated successful autonomous cattle feeding operation. However, there is considerable scope for future enhancement to improve system performance and functionality in large-scale farming environments. One major improvement is the integration of obstacle detection sensors such as ultrasonic or LiDAR sensors to improve autonomous navigation and avoid collisions during robot movement.

The system can also be enhanced by integrating GPS and IoT technologies for remote monitoring and real-time tracking of robot operation. A mobile application or cloud-based monitoring platform can be developed for controlling and monitoring the feeding system remotely. Increasing the feed hopper capacity and improving battery backup can further enhance continuous operation in large dairy farms.

The TensorFlow Lite AI model can be trained using larger and more diverse datasets to improve cattle detection accuracy under different environmental and lighting conditions. Future versions of the system may also include multiple feeding modes, automatic health monitoring of cattle, and advanced AI-based livestock behavior analysis.

Additionally, the robot can be upgraded with solar power support and advanced autonomous navigation techniques for fully automated smart farming applications. These future enhancements can improve the efficiency, reliability, and scalability of the automatic cattle feeding bot for modern precision livestock management systems.

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APPENDIX

Appendix section includes additional documents and supporting materials related to the project and technical activities undertaken during the academic period. These records are attached to provide a clear understanding of the work carried out, research contributions made, and participation in technical events. The appendix serves as supplementary evidence for the concepts, implementation, and outcomes discussed in the report.

The documents included in this section highlight the efforts taken towards research, innovation, and project presentation. They also demonstrate involvement in academic enrichment activities and technical learning experiences that contributed to the successful completion of the work. The following sections contain publication details, the detailed research paper, and the Technodea certificate for reference and documentation purposes.

APPENDIX A : Publication Details

International Journal of Scientific Research in Engineering and Management (IJSREM) is an international multidisciplinary journal that provides a platform for students, researchers, academicians, and professionals to publish innovative research work in the fields of engineering, science, and modern technology. The journal encourages the publication of technical research, project implementations, and emerging technological developments that contribute to academic learning and industrial advancement.

IJSREM follows a peer-review process to maintain the originality, quality, and authenticity of the published research papers. As an open-access journal, it allows researchers and readers from different parts of the world to access published articles freely, thereby promoting research collaboration and knowledge sharing. The journal also supports interdisciplinary research by publishing papers from various engineering and technological domains.

Publishing the paper titled “Automatic Cattle Feeding Bot” in IJSREM provided an opportunity to present innovative work related to computer vision, embedded systems, and robotic automation in smart farming applications. The research work focused on developing an autonomous livestock feeding robot capable of performing real-time cattle detection and automatic feed dispensing using Raspberry Pi 5, TensorFlow Lite, OpenCV, and robotic control technologies.

The publication helped improve technical writing skills, research presentation abilities, and understanding of the academic publication process. Publishing the work in IJSREM served as a valuable platform for showcasing the practical implementation and research contribution of the project in the field of smart livestock management and agricultural automation systems.

APPENDIX B : Detailed Paper

The Detailed technical paper prepared based on the implemented Project work is attached below



International Journal of Scientific Research in Engineering and Management (IJSREM)
Volume: 10 Issue: 05 | May - 2026 SJIF Rating: 8.659 ISSN: 2582-3930

Automatic Cattle Feeding Bot

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Abstract

Modern smart agriculture techniques have been much improved by the quick development of artificial intelligence, visual computing, and integrated automation technologies. This article describes the design and implementation of an artificial intelligence-driven automatic cattle feeding robot meant for effective cattle feeding operations. Through fast cattle identification and automated feed delivery systems, the proposed method tries to reduce human labor, improve feeding homogeneity, and enable self-operating farm management. A Raspberry Pi 5, the TensorFlow Lite AI framework, OpenCV for picture analysis, a USB webcam, an L298N motor driver, DC motors, and a servo motor-based feed dispensing unit make up this system. The USB webcam constantly records real-time video feed from the agricultural environment, and the captured photos are evaluated with OpenCV and TensorFlow Lite for cattle identification reasons. The robot stops and starts the servo system to distribute predetermined amounts of feed when cattle are precisely located with sufficient certainty. After the feed distribution, the robot returns to its autonomous navigation and maintains surveillance of its surroundings. Developed system efficiently demonstrated real-time cattle identification, self-guided robot navigation, accurate feed dispensing, and reduced human involvement throughout reviews. The suggested Automatic Cattle Feeding Bot is an affordable, intelligent, and portable answer for smart farming that fits right into the current livestock management and agricultural automation industries.

Index Terms – Artificial Intelligence (AI), Raspberry

Pi 5, TensorFlow Lite, OpenCV, Computer Vision, Smart Farming, Livestock Automation, Autonomous Robot, Cattle Detection, Feed Dispensing System, Embedded Systems, Agricultural Automation.

Introduction

Agriculture and livestock farming play a vital role in global food production and economic development. In dairy farms, proper feeding management is essential for maintaining cattle health, improving milk production, and increasing overall farm productivity. Traditional cattle feeding methods mainly depend on manual labor, where farmers are required to feed cattle regularly and continuously monitor feeding operations. These methods become difficult, labor-intensive, and time-consuming, particularly in farms with a large number of cattle. Irregular feeding schedules, improper feed distribution, and excessive feed wastage can negatively affect livestock health and farm efficiency. Therefore, there is an increasing demand for intelligent and automated systems capable of simplifying cattle feeding operations while reducing human effort.

Recent advancements in Artificial Intelligence (AI), computer vision, robotics, and embedded automation technologies have enabled the development of modern smart farming systems. Automated livestock management solutions using AI and image processing techniques are becoming increasingly popular in precision agriculture applications. Computer vision systems integrated with embedded platforms can identify cattle in real time using camera-based monitoring, thereby improving the accuracy and efficiency of feeding operations. Similarly, autonomous robotic systems can reduce farmer dependency on manual supervision and support continuous livestock monitoring inside farm environments.

The proposed Automatic Cattle Feeding Bot is developed as an intelligent and autonomous livestock feeding system capable of detecting cattle and dispensing feed automatically without continuous human intervention.



The system is implemented using Raspberry Pi 5, TensorFlow Lite AI model, OpenCV image processing, USB web camera, L298N motor driver, DC motors, and a servo motor-based feed dispensing mechanism. The USB web camera continuously captures real-time video frames from the surrounding environment, and the captured images are processed using OpenCV and TensorFlow Lite for cattle detection. When cattle are detected successfully with sufficient confidence, the robot automatically stops and activates the servo-controlled feed dispensing mechanism to release feed in controlled quantity. After feed dispensing is completed, the robot resumes autonomous movement and continues monitoring the environment.

The integration of Artificial Intelligence, computer vision, embedded systems, and robotic automation technologies enables the proposed system to perform intelligent livestock feeding operations efficiently and accurately. The developed system reduces manual labor, minimizes feed wastage, improves feeding consistency, and supports autonomous farm management. The Automatic Cattle Feeding Bot provides a cost-effective, portable, and intelligent smart farming solution suitable for modern livestock management and agricultural automation applications.

LITERATURE REVIEW

Recent advancements in Artificial Intelligence (AI), computer vision, robotics, and embedded systems have significantly influenced the development of smart farming and autonomous livestock management technologies. Researchers have increasingly focused on improving agricultural automation systems using intelligent monitoring, image processing, and robotic control techniques. However, many existing systems mainly concentrate on monitoring or feeding operations individually and do not provide a complete integrated autonomous livestock feeding solution.

Umapathi et al. [1] proposed an IoT-based smart livestock feeding system using embedded automation technologies for improving feeding efficiency in dairy farms. The system utilized automated feed control mechanisms and monitoring technologies to reduce manual labor. The study demonstrated that IoT-based smart feeding systems can improve farm productivity and feeding consistency. However, the proposed work mainly focused on IoT-based monitoring and did not include AI-based cattle detection using computer vision techniques.

Giannone et al. [2] presented a computer vision-based cattle monitoring system for real-time livestock management applications. The system used camera-based image acquisition and Artificial Intelligence techniques for cattle identification and monitoring. The research demonstrated that image processing and deep learning models can accurately identify cattle in farm environments. However, the proposed work mainly focused on cattle monitoring and did not integrate autonomous robotic feed dispensing operations.

Souza et al. [3] discussed precision feeding techniques for dairy farms and highlighted the importance of controlled

feed dispensing in improving cattle productivity and reducing feed wastage. The study explained that automated feeding technologies can improve livestock health, feeding efficiency, and milk production while reducing human intervention in farm operations. The work emphasized the growing importance of smart feeding technologies in modern livestock management systems.

Tinoco et al. [4] proposed a multi-sensory livestock monitoring system integrated with wireless communication and embedded automation technologies. The system used various sensing and monitoring techniques for observing cattle activities and improving smart farming operations. The study demonstrated that intelligent monitoring systems can support precision livestock farming and reduce manual supervision requirements. However, the system did not include AI-based image classification and autonomous robotic navigation features.

Chelotti et al. [5] reviewed automated sensing and monitoring techniques used in livestock farming applications. The study analyzed various image processing methods, sensing technologies, and embedded automation systems used for monitoring cattle behavior and feeding operations. The paper concluded that Artificial Intelligence and computer vision technologies have significant potential in improving agricultural automation and smart livestock management systems.

Siraskar et al. [6] developed an automated cattle feeding mechanism using embedded systems and motor control techniques. The proposed system reduced manual feeding effort and improved feeding consistency through automated feed dispensing operations. The research highlighted the importance of low-cost automation technologies for dairy farms and demonstrated the practical application of embedded systems in smart farming applications.

Google Developers [7] presented TensorFlow Lite as a lightweight Artificial Intelligence framework suitable for embedded systems and edge AI applications. TensorFlow Lite enables fast and efficient real-time image classification with low memory consumption, making it highly suitable for Raspberry Pi-based smart automation systems. The framework supports real-time object detection and intelligent decision-making operations in embedded AI applications.

From the literature survey, it is observed that most existing systems mainly focus on livestock monitoring, IoT-based automation, or feed dispensing operations separately. Very few systems combine Artificial Intelligence, computer vision, autonomous robotic navigation, and automatic feed dispensing into a single integrated smart farming solution. Therefore, the proposed Automatic Cattle Feeding Bot is developed by integrating TensorFlow Lite AI model, OpenCV image processing, Raspberry Pi 5, autonomous robotic movement, and servo-based feed dispensing mechanism for intelligent and efficient livestock feeding operations.

METHODOLOGY

The methodology adopted for the Automatic Cattle Feeding Bot focuses on developing an intelligent



and autonomous livestock feeding system using Artificial Intelligence, computer vision, embedded systems, and robotic automation technologies. The proposed system is designed to automate cattle feeding operations by detecting cattle in real time and dispensing feed automatically without requiring continuous human supervision. The methodology mainly involves image acquisition, AI-based cattle detection, autonomous robot movement, and servo-controlled feed dispensing operations.

The complete system architecture consists of Raspberry Pi 5, USB web camera, TensorFlow Lite AI model, OpenCV image processing, L298N motor driver, DC motors, servo motor, feed hopper mechanism, and power supply unit. The Raspberry Pi acts as the central processing and control unit of the entire system. The camera continuously captures real-time images from the surrounding environment and transfers them to the Raspberry Pi for processing and cattle detection operations.

1. Image Acquisition and Preprocessing

The USB web camera continuously captures real-time video frames from the farm environment while the robot moves autonomously. These captured image frames are transferred to the Raspberry Pi for processing. OpenCV image processing techniques are used for preprocessing operations such as image resizing, normalization, filtering, and feature extraction. The preprocessing stage improves image quality and prepares the frames for efficient AI-based cattle detection.

2. AI-Based Cattle Detection

The processed image frames are analyzed using the TensorFlow Lite Artificial Intelligence model trained for cattle detection. The AI model compares the extracted image features with trained cattle image datasets and predicts whether cattle are present in the captured frame. The confidence score generated by the model is compared with a predefined threshold value. If the confidence value exceeds the threshold limit, cattle detection is confirmed successfully.

3. Autonomous Robot Navigation

During normal operation, the Raspberry Pi continuously controls the DC motors through the L298N motor driver to move the robot autonomously inside the farm environment. The robot continuously scans the surroundings for cattle presence while navigating through the farm area. If cattle are not detected, the robot continues forward movement without interruption.

4. Feed Dispensing Operation

When cattle are detected successfully, the Raspberry Pi immediately stops the robot movement by controlling the motor driver. After stopping near the cattle, the Raspberry Pi activates the servo motor connected to the feed dispensing mechanism. The servo motor rotates and opens the flap door of the feed hopper, allowing feed to be dispensed automatically in controlled quantity. After feed dispensing for a fixed duration, the

servo motor returns to its original position and closes the feed outlet flap automatically.

5. Continuous Monitoring and Return Operation

After completing feed dispensing, the robot resumes autonomous movement and continues monitoring the surrounding environment for additional cattle detection. A cooldown mechanism is implemented to prevent repeated feeding of the same cattle within a short duration. The system continuously performs cattle detection, robot navigation, and feed dispensing operations without requiring human intervention. The implemented methodology enables intelligent livestock feeding, reduces manual labor, minimizes feed wastage, and improves automation efficiency in smart farming applications.

OBJECTIVES

1. Automated Cattle Feeding:

The main objective of the proposed system is to develop an autonomous robotic feeding system capable of dispensing feed automatically to cattle without requiring continuous human supervision. The system is designed to reduce the manual labor involved in traditional livestock feeding operations and improve feeding efficiency in dairy farms.

2. AI-Based Cattle Detection:

Another important objective is to implement a real-time cattle detection system using TensorFlow Lite Artificial Intelligence model and OpenCV image processing techniques. The camera-based detection system continuously monitors the farm environment and identifies cattle accurately before initiating the feed dispensing operation.

3. Autonomous Robot Navigation:

The project aims to design a robotic platform capable of moving autonomously inside the farm environment using DC motors and motor driver control. The robot continuously travels through the farm while monitoring cattle presence and performing feeding operations automatically.

4. Controlled Feed Dispensing:

The proposed system also aims to develop a servo motor-based feed dispensing mechanism that releases feed in controlled quantity whenever cattle are detected successfully. This controlled dispensing operation helps reduce feed wastage and ensures proper feed distribution for livestock management.

PROBLEM STATEMENT

1. Manual and Labor-Intensive Feeding Operations:

Traditional cattle feeding methods require continuous human effort and regular monitoring. Farmers must manually feed cattle at scheduled intervals every day, making the process time-consuming and labor-intensive, especially in farms with a large number of cattle.

2. Irregular Feeding and Feed Wastage:

Improper feeding schedules and inconsistent feed distribution may negatively affect cattle health, growth, and milk production. Manual feeding methods can also lead to

excessive feed wastage and inefficient feed utilization.

3. Lack of Intelligent Cattle Detection:

Most conventional automated feeding systems operate using fixed timers or predefined schedules without verifying the actual presence of cattle. This reduces feeding accuracy and may result in unnecessary feed dispensing operations.

4. Limited Integration of AI and Automation:

Many existing livestock monitoring systems mainly focus on IoT monitoring or simple feed dispensing mechanisms separately. Very few systems combine Artificial Intelligence, computer vision, autonomous navigation, and automatic feed dispensing into a single integrated smart farming solution.

5. Difficulty in Continuous Livestock Monitoring:

Continuous monitoring of cattle inside large farm environments is difficult for farmers and requires significant manpower. Lack of intelligent monitoring systems may reduce operational efficiency and increase dependency on manual supervision.

SOLUTION OF THE PROBLEM

1. Automated Cattle Feeding Operation:

The proposed Automatic Cattle Feeding Bot automates the cattle feeding process using Raspberry Pi 5, DC motors, and a servo-controlled feed dispensing mechanism. The robot moves autonomously inside the farm environment and dispenses feed automatically without requiring continuous human intervention.

2. AI-Based Real-Time Cattle Detection:

The system uses a USB web camera along with OpenCV and TensorFlow Lite AI model for real-time cattle detection. The AI model continuously analyzes captured image frames and confirms cattle presence before activating the feed dispensing mechanism.

3. Controlled Feed Dispensing Mechanism:

A servo motor-controlled feed hopper is used to dispense feed in controlled quantity only when cattle are detected successfully. This minimizes feed wastage and improves feeding consistency in dairy farms.

4. Integration of AI and Robotic Automation:

The proposed system combines Artificial Intelligence, computer vision, embedded systems, and autonomous robotic movement into a single smart farming platform. This integration improves automation efficiency and enables intelligent livestock feeding operations.

5. Continuous Autonomous Monitoring:

The robot continuously monitors the surrounding farm environment using camera-based image acquisition and autonomous movement control. This reduces human supervision requirements and improves operational efficiency in livestock management applications.

BLOCK DIAGRAM

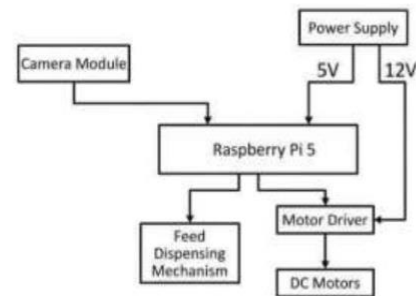


Figure 1: Block Diagram of the proposed system

The block diagram of the Automatic Cattle Feeding Bot represents the overall architecture and working connection between the major hardware components used in the system. The system mainly consists of a Camera Module, Raspberry Pi 5, Motor Driver, DC Motors, Feed Dispensing Mechanism, and Power Supply unit. All these components work together to perform autonomous cattle detection and automatic feed dispensing operations.

The **Camera Module** is connected directly to the Raspberry Pi 5 and continuously captures real-time video frames from the surrounding farm environment. The captured images are transferred to the Raspberry Pi for image processing and cattle detection using OpenCV and TensorFlow Lite Artificial Intelligence model. The camera acts as the vision system of the robot and enables real-time monitoring of cattle presence. The **Raspberry Pi 5** serves as the central processing and control unit of the entire system. It receives image data from the camera module and processes the captured frames using AI and computer vision techniques. Based on the cattle detection result, the Raspberry Pi controls the movement of the robot and activates the feed dispensing mechanism automatically. The Raspberry Pi also sends control signals to the motor driver for controlling the DC motors.

The **Motor Driver** acts as an interface between the Raspberry Pi and the DC motors. Since the GPIO pins of the Raspberry Pi cannot provide sufficient current required for motor operation, the motor driver amplifies the control signals and supplies the required power to the motors. The motor driver controls the direction and movement of the robot based on commands received from the Raspberry Pi.

The **DC Motors** are connected to the motor driver and are responsible for autonomous movement of the robot inside the farm environment. During normal operation, the motors move the robot continuously for cattle monitoring. When cattle are detected successfully, the Raspberry Pi stops the motors and initiates the feed dispensing process.

The **Feed Dispensing Mechanism** is controlled by the Raspberry Pi using a servo motor arrangement. When cattle are detected, the Raspberry Pi activates the servo motor, which opens the feed outlet flap and dispenses feed automatically in controlled quantity. After feed dispensing is completed, the flap closes automatically and the robot resumes movement.

The **Power Supply** provides electrical power to all hardware components used in the system. A regulated **5V supply** is

provided to the Raspberry Pi 5 for stable operation of electronic components, while a **12V supply** is provided to the motor driver and DC motors for autonomous robot movement. The dual-power arrangement ensures efficient and reliable operation of the entire system.

Thus, the block diagram illustrates the integration of Artificial Intelligence, image processing, embedded systems, and robotic automation technologies for intelligent and autonomous livestock feeding operations

SYSTEM DESIGN

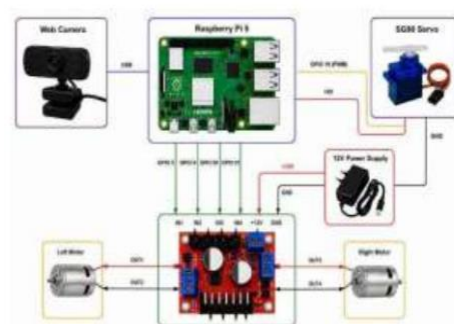


Figure 2: Circuit Design

The Automatic Cattle Feeding Bot integrates Raspberry Pi 5, USB web camera, TensorFlow Lite AI model, OpenCV image processing, L298N motor driver, DC motors, servo-based feed dispensing mechanism, and power supply unit into a unified architecture, where each component communicates efficiently to perform real-time cattle detection, autonomous robot navigation, and intelligent automatic feed dispensing operations.

1. USB Web Camera Module:

Captures real-time video frames from the surrounding farm environment and transfers the captured images to the Raspberry Pi 5 for cattle detection and image processing operations.

2. Raspberry Pi 5:

Acts as the central processing and control unit of the system, performing image processing, TensorFlow Lite AI execution, motor control, and automatic feed dispensing operations.

3. L298N Motor Driver:

Amplifies control signals received from the Raspberry Pi and supplies sufficient current and voltage required for operating the DC motors efficiently.

4. DC Gear Motors:

Drive the robot autonomously inside the farm environment and enable continuous cattle monitoring and navigation operations.

5. SG90 Servo Motor:

Controls the feed dispensing mechanism by rotating the flap

door of the feed hopper for automatic feed delivery in controlled quantity.

6. Feed Dispensing Mechanism:

Dispenses cattle feed automatically whenever cattle are detected successfully by the Artificial Intelligence model.

7. 12V Power Supply:

Provides electrical power to the motor driver, DC motors, Raspberry Pi, and other hardware components used in the system.

8. Robotic Chassis:

Acts as the structural frame of the robot and supports all hardware components required for autonomous movement and feeding operations.

9. Wheels:

Attached to the DC motors and enable smooth movement of the robot across the farm environment during monitoring and feeding operations.

HARDWARE & SOFTWARE REQUIREMENTS

1. Hardware Requirements:

- Raspberry Pi 5
- USB Web Camera Module
- L298N Motor Driver
- DC Gear Motors
- SG90 Servo Motor
- Feed Dispensing Mechanism
- Robotic Chassis
- 12V Power Supply

2. Software Requirements:

- Raspberry Pi OS
- Python
- OpenCV
- TensorFlow Lite

ADVANTAGES

1. Reduces Manual Labour:

The Automatic Cattle Feeding Bot automates the entire cattle feeding process, reducing the need for farmers to manually carry and distribute feed regularly. This minimizes human effort and makes livestock management easier, especially in farms with a large number of cattle.

2. Reduced Feed Wastage:

Feed is dispensed only when cattle are detected successfully by the TensorFlow Lite AI model. This controlled feed dispensing mechanism minimizes unnecessary feed wastage and improves feed utilization efficiency.



3. Consistent Feeding:

The system dispenses feed in controlled quantities whenever cattle are detected successfully. This helps maintain proper feeding consistency and ensures that cattle receive feed at regular intervals, improving livestock health and farm productivity.

4. Autonomous Operation:

The robot operates autonomously using camera-based cattle detection and embedded control systems. It continuously monitors the farm environment, detects cattle in real time, and performs feed dispensing operations automatically without requiring continuous human intervention.

5. Reduced Human Supervision:

The proposed system reduces the requirement for continuous monitoring by farmers. The robot performs cattle detection, navigation, and feed dispensing operations independently, allowing farmers to focus on other farm management activities.

6. Improved Smart Farming Efficiency:

The integration of Artificial Intelligence, computer vision, embedded systems, and robotic automation technologies improves operational efficiency in modern smart farming applications and supports intelligent livestock management.

RESULTS & DISCUSSIONS

The proposed Automatic Cattle Feeding Bot was successfully developed and tested using Raspberry Pi 5, TensorFlow Lite, OpenCV, DC motors, L298N motor driver, USB web camera, and SG90 servo motor. Experimental testing confirmed that the system was capable of performing intelligent cattle detection, autonomous navigation, and automatic feed dispensing operations effectively.

During operation, the USB web camera continuously captured live video frames from the farm environment. These frames were processed by the Raspberry Pi using OpenCV image processing techniques and the TensorFlow Lite AI model. The trained model successfully identified cattle in real time under normal environmental conditions with reliable detection performance.

The robotic movement system operated smoothly throughout the testing phase. The DC motors controlled through the L298N motor driver enabled continuous forward movement of the robot while scanning the surroundings for cattle presence. The movement control mechanism also allowed immediate stopping of the robot whenever cattle were detected successfully.

Once cattle were identified with sufficient confidence, the Raspberry Pi activated the SG90 servo motor connected

to the feed dispensing mechanism. The servo motor opened the feed outlet flap and released feed in controlled quantity for a fixed duration. After dispensing was completed, the flap closed automatically and the robot resumed monitoring operation.

The implemented cooldown mechanism helped avoid repeated feeding of the same cattle within a short time interval. This improved feeding efficiency and reduced unnecessary feed wastage during continuous operation. The return movement mechanism also operated successfully, allowing the robot to move back approximately to its starting position after completing the feeding cycle.

The developed system demonstrated reliable autonomous operation with minimal human intervention. The integration of Artificial Intelligence, computer vision, embedded control, and robotic automation technologies improved livestock feeding efficiency and reduced manual labor requirements in smart farming applications.

Overall, the Automatic Cattle Feeding Bot provided an intelligent, cost-effective, and portable solution for modern livestock management and agricultural automation system.

CONCLUSION

This work presented the development of an intelligent Automatic Cattle Feeding Bot designed for modern smart farming applications. The proposed system combined Artificial Intelligence, image processing, embedded control, and robotic automation to perform autonomous cattle feeding operations with reduced human involvement.

The implemented system utilized Raspberry Pi 5, TensorFlow Lite, OpenCV, USB web camera, L298N motor driver, DC motors, and servo-based feed dispensing mechanism for real-time cattle detection and automated feeding. Experimental testing confirmed that the robot was capable of detecting cattle accurately, navigating autonomously inside the farm environment, and dispensing feed in controlled quantity whenever livestock presence was identified successfully.

The automated feeding mechanism reduced manual labor requirements and improved feeding consistency compared to conventional livestock feeding methods. The use of AI-based cattle detection also minimized unnecessary feed wastage by ensuring that feed was dispensed only when cattle were present near the robot.

Overall, the developed Automatic Cattle Feeding Bot demonstrated reliable performance and effective integration of Artificial Intelligence and robotic automation technologies for livestock management applications. The proposed system provides a practical, cost-effective, and scalable solution for smart dairy farming and agricultural automation environments.



FUTURE SCOPE

Obstacle detection sensors can be integrated to improve autonomous navigation and avoid collisions during robot movement. IoT and cloud connectivity can be added for remote monitoring and control of feeding operations. The TensorFlow Lite AI model can be trained using larger datasets to improve cattle detection accuracy under different environmental conditions. In future, the system can also be upgraded with cattle health monitoring, GPS tracking, and solar power support for advanced smart farming applications.

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APPENDIX C : KSCST Submission

The section contains the documents related to the submission of the project to the Karnataka State Council for Science and Technology (KSCST). The submission included the project abstract, objectives, methodology, implementation, and expected outcomes. It provided valuable experience in technical documentation, research, teamwork, and project presentation while encouraging innovation and practical problem-solving skills. The following pages contain the KSCST submission documents for reference and record purposes.



KARNATAKA STATE COUNCIL FOR SCIENCE AND TECHNOLOGY Indian Institute of Science campus, Bengaluru

Telephone: 080 – 23341652, 23348848 || Email: spp@kscst.org.in
Website: www.kscst.org.in/spp.html or https://kscst.karnataka.gov.in/en

FORMAT FOR STUDENT PROJECT PROPOSAL FOR THE 49th SERIES OF STUDENT PROJECT PROGRAMME

1.	Name of the College: SRI SIDDHARTHA INSTITUTE OF TECHNOLOGY, TUMAKURU
2.	Project Title: Automatic Cattle Feeding Bot
3.	Branch: Electronics and Communication Engineering
4.	Theme: Agricultural Technology: Intelligent Robotic Systems for Automated Cattle Feeding.
5.	Name of project guide: 1. Name: Dr.Sreerangappa M Email id: sreerangappa@ssit.edu.in Contact No: 9880581860
6.	Name of Team Members: Name: RAJESHWARI B B USN No.: 22EC075 Email id: rajeshwaribiradar937@gmail.com Mobile No: 7892216493  Name: S G SHILPA USN No.: 22EC078 Email id: sgshilpasharu@gmail.com Mobile No.: 8073805649 

	Name: SHREEVATHSA USN No.: 22EC091 Email id: Shreevathsa804@gmail.com Mobile No.: 9019691526  Name: T PRAVEEN KUMAR USN No.: 22EC101 Email id: tpraveenkumar798@gmail.com Mobile No.: 9390920349 
7.	Team Leader of the Project: Name: SHREEVATHSA USN No.: 22EC091 Email id: Shreevathsa804@gmail.com Mobile No.: 9019691526
8.	Processing Fee Details (Through Online Payment only): (processing fee of Rs. 1180/-) Payment details enclosed in the last page of the proposal
9.	Date of commencement of the Project: 22/09/2025
10.	Probable date of completion of the project: 1/04/2026

APPENDIX D : Technodea Certificate

Technodea-2026 was a national-level project exhibition and competition conducted on 8th and 9th May 2026 at Sri Siddhartha Institute of Technology, Tumakuru. The event provided an opportunity for us to showcase innovative technical projects, exchange ideas, and enhance their knowledge through academic and technical interaction.



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(A constituent college of Sri Siddhartha Academy of Higher Education)
(Deemed to be university under section 3 of UGC act 1956, Approved by AICTE, Accredited by NBA and NAAC with 'A+' Grade.)

NATIONAL LEVEL PROJECT EXHIBITION & COMPETITION

Technodea -2026

CERTIFICATE OF PARTICIPATION

This is to certify that Mr./ Ms. Shreevathsa
of SSIT from Dept. of Electronics and Communication Engineering
has participated in "Technodea", A National level Project Competition under
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APPENDIX E : SDG Mapping



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Sustainable Development Goals (SDGs)
Academic year: 2025-26

Title of the work: “AUTOMATIC CATTLE FEEDING BOT”

Objectives of the project (Major):

1. To develop an autonomous robotic feeding system capable of dispensing feed automatically to cattle with minimal human intervention.
2. To implement real-time cattle detection using TensorFlow Lite and OpenCV image processing techniques for intelligent feeding operations.
3. To design a robotic platform capable of autonomous movement, controlled feed dispensing, and continuous livestock monitoring for smart farming applications.
4. To reduce feed wastage and improve feeding efficiency using camera based cattle identification and automated feeding mechanisms.

17 Sustainable Development Goals (SDGs) are:

Goal No.	Sustainable Development Goals (SDGs)	Level
1	No poverty	-
2	Zero hunger	2
3	Good health and well-being	-
4	Quality education	-
5	Gender equality	-
6	Clean water and sanitation	-
7	Affordable and clean energy	-
8	Decent work and economic growth	-
9	Industry, innovation and infrastructure	3
10	Reduced inequalities	-
11	Sustainable cities and communities	-
12	Responsible consumption and production	2
13	Climate action	-
14	Life below water	-
15	Life on land	2
16	Peace, justice, and strong institutions	-
17	Partnerships for the goals	-

Levels: High-3, Medium-2, Low-1

Outcomes of the work:

1. The automatic cattle feeding bot successfully performs real-time cattle detection and automatic feed dispensing using TensorFlow Lite, OpenCV, and Raspberry Pi 5.
2. The developed robotic platform reduces manual labor and improves feeding efficiency through autonomous navigation and controlled feed dispensing operations.
3. The implemented cooldown mechanism minimizes unnecessary feed wastage by preventing repeated feeding of the same cattle within short durations.
4. The project demonstrates effective integration of computer vision, embedded systems, and robotic automation technologies for smart livestock